The standard integral as a rescaled fluctuation function: $\int_0^\infty u^h e^{-\beta n u^{2k} + \beta\sqrt{n}\, u^k a}\, du = \frac{1}{2k} n^{-(h+1)/(2k)} S_{(h+1)/(2k)}(a)$

Claude, with Daniel Murfet

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Abstract

The first foothold into grammar §4.2 (the Taylor-tree / standard-integral machinery), bridging it to the §4.1 fluctuation function now fully formalised. We prove that the one-dimensional constant-coefficient population core of the per-chart standard integral (`eq:per_chart_integral`) — $d = 1$, $\xi \equiv a$, $\eta \equiv 1$, over the full half-line — is exactly a rescaled fluctuation function, via the change of variables $t = nu^{2k}$. This identifies S_λ as the exact Mellin kernel of the standard integral and pins down the leading candidate exponent $(h+1)/(2k)$ of $\Lambda(h, k)$ (`eq:candidateexponents`). Zero `sorry`/`axiom`.

1 Setting

The standard integral $Z(\beta, n; \xi, \eta) = \int_{[0,b]^d} u^h e^{-\beta n u^{2k} + \beta\sqrt{n}\, u^k \xi(u)} \eta(u)\, du$ is the object whose full asymptotic expansion is `thm:TaylorTree`. That theorem is a heavy multivariate result (Dirac state densities, Mellin transforms in $\mathbb{C}$, multi-index tree combinatorics, meromorphic continuation) — not a single tide. But its *analytic core* in one dimension, with constant coefficients and no boundary truncation, is elementary and directly consumes the §4.1 fluctuation function. Isolating that core is the right first step.

2 The thing formalised

```
theorem standardIntegral1D_eq_fluctuation (  n a :  ) (h k :  )
    (hn : 0 < n) (hk : 0 < k) :
    (  u in Set.Ioi (0: ),
        u ^ h * Real.exp (- *n*u^(2*k) +  *Real.sqrt n*u^k*a))
      = (1 / (2*(k: ))) * n ^ (-(((h: )+1)/(2*k)))
        * fluctuation  (((h: )+1)/(2*k)) a
```

The order of the fluctuation function produced is $\lambda = (h+1)/(2k)$ — precisely the starting point of the i-th arithmetic progression in $\Lambda(h, k) = \bigcup_i \big((h_i{+}1)/(2k_i) + (1/2k_i)\mathbb{N}\big)$. The theorem needs no sign or size assumption on β: the substitution is algebraic and carries β along untouched, so the result holds for all real β.

3 Proof strategy

The substitution $t = nu^{2k}$ factors as a scaling by n and a power $u \mapsto u^{2k}$, matched to the two Mathlib change-of-variables lemmas (the same pair used by `fluctuation_eq_kernel`, here at general $p = 2k$ instead of $p = 2$):

- `integral_comp_mul_left_Ioi`: $\int_0^\infty G(nr)\,dr = n^{-1}\int_0^\infty G(t)\,dt$, so $S_\lambda = \int G = n\int G(n\,\cdot)$;
- `integral_comp_rpow_Ioi_of_pos`: $\int_0^\infty (px^{p-1})\,G(nx^p)\,dx = \int_0^\infty G(nr)\,dr$.

The integrand $(px^{p-1})G(nx^p)$ then collapses pointwise (for $x>0$) to $2k\,n^{\lambda-1}\,x^h e^{-\beta n x^{2k}+\beta\sqrt{n}x^k a}$: the x-powers merge as $x^{(p-1)+p(\lambda-1)} = x^h$ (the exponent arithmetic $(2k-1)+((h{+}1)-2k)=h$ is where $\lambda=(h{+}1)/(2k)$ is *used*), and $\sqrt{nx^{2k}} = \sqrt{n}\,x^k$. Pulling out the constant $2k\,n^{\lambda-1}$ gives $S_\lambda = 2k\,n^\lambda Z$, and dividing yields the stated form.

4 Roadblocks and resolutions

- The power bookkeeping mixes `npow` (u^{2k}, u^k with $k:\mathbb{N}$) and `rpow` (x^p, $x^{\lambda-1}$). Bridge with `Real.rpow_natCast` ($x^{(m:\mathbb{N})} = x^m$) and set $p = ((2k:\mathbb{N}):\mathbb{R})$ so $x^p = x^{2k}$ is one `rpow_natCast` away; the radical uses $x^{2k} = (x^k)^2$ (`pow_mul` + `Nat.mul_comm`) then `Real.sqrt_sq`.
- `ring` cannot see `rpow` exponent addition, so merge the two x-powers by `Real.rpow_add` and prove the resulting exponent equals $(h:\mathbb{R})$ by `field_simp; ring` (needs $2k\neq 0$); likewise $n\cdot n^{\lambda-1} = n^\lambda$ and $n^{-\lambda}n^\lambda = 1$ each need an explicit `rpow_add`/`rpow_zero` rewrite before `ring` finishes the surrounding scalar algebra.
- The exponent-term reorder $-\beta(nx^{2k}) + \beta a(\sqrt{n}x^k) = -\beta n x^{2k} + \beta\sqrt{n}x^k a$ cannot go under `exp` via `ring`; state it as a separate `have` closed by `ring` and `rw` it into the exponent.
- Order the integrand rewrites so `hnpow` (which consumes $(nx^p)^{\lambda-1}$) fires before `hxp` ($x^p \to x^{2k}$) rewrites the remaining x^p inside the exponent; the pattern x^p never matches x^{p-1} or $x^{p(\lambda-1)}$, so the rewrites stay surgical.

5 What was Mathlib, what was new

Mathlib supplied the change-of-variables lemmas and the `rpow` calculus (`Real.rpow_natCast`, `rpow_add`, `rpow_mul`, `mul_rpow`, `sqrt_mul`, `sqrt_sq`). New is the identification itself — §4.1's fluctuation function *is* the state-density Mellin kernel of §4.2's standard integral — and the exponent arithmetic exhibiting $(h+1)/(2k)$.

6 Lessons

`fluctuation_eq_kernel` was a direct template at general p: when a second change of variables has the same scaling-then-power shape, reuse the lemma pair and only redo the pointwise integrand collapse. The empirical pre-check (relative error $\sim 10^{-13}$ across four parameter sets) confirmed both the constant $1/(2k)$ and the exponent $-(h+1)/(2k)$ before any Lean was written.

7 Follow-ups

Natural next pieces toward `thm:TaylorTree`: the boundary-tail remainder Δ over $[b,\infty)$ (exponentially small), the general-coefficient expansion of $e^{\beta\sqrt{n}u^k(\xi(u)-\xi(0))}$ into the tree terms $\xi_{\mathbf{n},p}$ (`eq:flucttreeterms`), and the Mellin/meromorphic-continuation machinery for the state density — all substantially heavier and multivariate.